

Adaptive Authentication with Behavioral Biometrics

Project Planning & Management Document

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1. Project Proposal

1.1 Project Overview

The Adaptive Authentication with Behavioral Biometrics project proposes the design and implementation of a next-generation security system for web and mobile applications. Current authentication paradigms rely overwhelmingly on static, knowledge-based credentials (passwords, PINs), which are vulnerable to a wide range of cyber threats. This project introduces a continuous, behavior-driven authentication layer that operates transparently in the background, significantly reducing the risk of unauthorized account access — even in scenarios where login credentials have been compromised.

The system continuously analyzes unique behavioral signals — including keystroke dynamics, mouse movement patterns, and touchscreen gesture characteristics — to construct a personalized behavioral fingerprint for each user. Deviations from this fingerprint, detected via machine learning anomaly detection models, trigger graduated security responses proportional to the assessed risk level.

1.2 Project Objectives

- Design and implement a real-time behavioral biometrics data collection engine compatible with web browsers and mobile platforms.
- Develop machine learning-based anomaly detection models (One-Class SVM and Autoencoder) to establish and validate unique user behavioral fingerprints.
- Build a risk-scoring engine with configurable thresholds that triggers adaptive responses: step-up authentication or automatic session termination.
- Deliver a RESTful backend service and administrative dashboard enabling system integration, monitoring, and configuration.
- Validate system performance against defined KPIs including detection accuracy, false positive rate, and system uptime.
- Produce comprehensive technical documentation: system design, API reference, testing reports, and deployment guides.

1.3 Project Scope

In Scope

- Browser-based JavaScript SDK for keystroke, mouse, and touch event collection.
- .Net Core Web API: Auth Service, Behavior Collector, ML Inference Engine, Admin Service.
- One-Class SVM and Autoencoder model training, validation, and serving pipelines.
- Microsoft SQL server for database schema design, implementation, and migration scripts.
- React.js administrative dashboard with real-time anomaly monitoring.
- RESTful API with OpenAPI 3.0 documentation.
- Unit, integration, and user acceptance testing.

Out of Scope

- Hardware biometric devices (fingerprint readers, facial recognition cameras).
- Native iOS / Android SDK development (React Native mobile SDK is in scope).
- Integration with specific third-party identity providers (OAuth / SAML federation) beyond basic JWT compatibility.
- GDPR/compliance legal certification (compliance-by-design principles are followed but formal certification is not part of this project).

2. Project Plan

2.1 Project Timeline & Gantt Chart

The project is organized into four sequential phases spanning 20 weeks. Each phase builds upon the deliverables of the preceding one, with integration and testing activities overlapping development tasks to enable continuous validation.

Phase	Name	Weeks	Key Activities
Phase 1	Initiation & Design	W1 – W4	Proposal, requirements, system analysis, architecture design, ERD.
Phase 2	Core Development	W5 – W14	Database, backend APIs, behavioral SDK, ML model development & training.
Phase 3	Integration & Testing	W11 – W15	Frontend dashboard, API integration, security testing, performance testing.
Phase 4	Deployment & Closure	W16 – W20	UAT, production deployment, documentation, final presentation.

Table 2.1 — Project Phases Summary

Gantt Chart — 20-Week Project Schedule

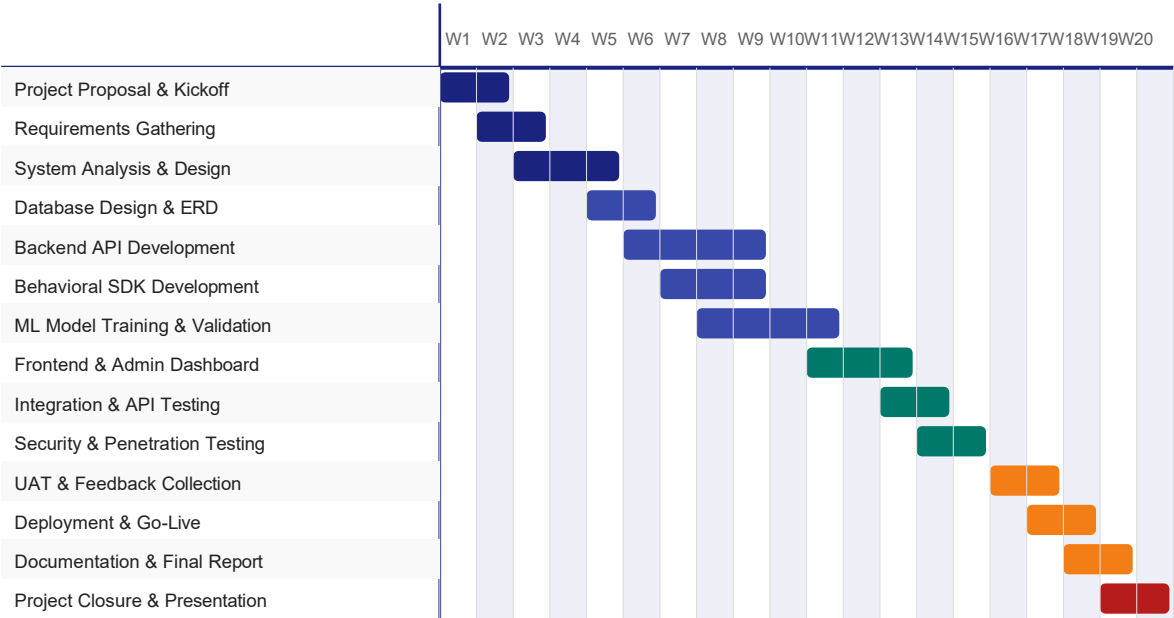
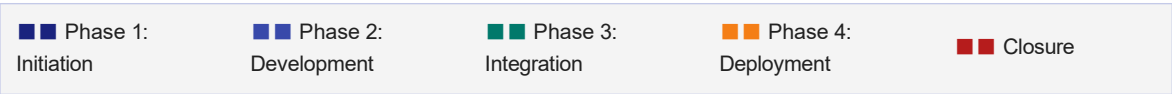


Figure 2.1 — Project Gantt Chart (W = Week)



2.2 Milestones & Deliverables

Milestone	Deliverable(s)	Target Week	Status
M1 — Project Kickoff	Approved project proposal, signed charter	W1	Planned
M2 — Design Complete	System analysis document, ERD, architecture diagram	W4	Planned
M3 — Database Ready	Implemented & migrated PostgreSQL schema	W6	Planned
M4 — Backend API v1.0	Auth, Collector, ML, Admin services deployed to staging	W10	Planned
M5 — ML Model Validated	Trained model achieving target FPR < 5%, TPR > 92%	W12	Planned
M6 — Frontend Complete	Admin dashboard deployed; all UI screens implemented	W13	Planned
M7 — Integration Complete	End-to-end test suite passing; security scan clean	W15	Planned
M8 — UAT Sign-Off	UAT report with stakeholder approval	W17	Planned
M9 — Production Deployment	Live system deployed; monitoring dashboards active	W18	Planned
M10 — Project Closure	Final documentation, presentation, lessons learned report	W20	Planned

Table 2.2 — Project Milestones & Deliverables

2.3 Resource Allocation

Resource	Type	Allocation	Phase(s)	Notes
Team Lead / PM	Human	100%	All	Oversees planning, reporting, stakeholder communication.
Backend Developer (x2)	Human	100%	2,3,4	FastAPI services, Kafka integration, database.
ML Engineer	Human	100%	2,3	Model development, training pipeline, inference engine.
Frontend Developer	Human	100%	2,3	React.js admin dashboard, behavioral SDK (JS).
QA / Security Eng.	Human	100%	3,4	Testing, penetration testing, CI/CD validation.
Dev. Laptops (x5)	Hardware	Dedicated	All	Min. 8 GB RAM, 8-core CPU for ML training.

Resource	Type	Allocation	Phase(s)	Notes
Project Mgmt. Tool	Software	Licensed	All	Jira / GitHub Projects for task tracking.
Communication Tool	Software	Licensed	All	Slack for team comms; Confluence for documentation.

Table 2.3 — Resource Allocation Plan

3. Task Assignment & Roles

3.1 Team Structure

The project team consists of five members operating under an Agile-influenced structure with two-week sprint cycles. Roles are defined to ensure clear accountability while allowing collaborative overlap during integration phases.

Role	Member	Primary Responsibilities	Key Deliverables
Project Manager / Tech Lead	Member A	Sprint planning, stakeholder reporting, architecture decisions, code review.	Project plan, architecture diagram, sprint reports, final presentation.
Backend Developer 1	Member B	Auth Service, session management, API Gateway configuration, JWT/security.	Auth API, session management module, API documentation.
ML Engineer	Member D	Feature engineering, One-Class SVM, Autoencoder training, ML inference API.	Trained models, model validation report, ML inference service, performance benchmarks.
Frontend Developer / QA	Member E	React.js admin dashboard, JavaScript behavioral SDK, testing (unit, integration, UAT).	Admin dashboard, JS SDK, test plans, UAT report, UI/UX documentation.

Table 3.1 — Team Roles & Responsibilities

3.2 RACI Matrix

The RACI matrix defines Responsible (R), Accountable (A), Consulted (C), and Informed (I) assignments across key project activities.

Activity	PM/Lead	Backend 1	ML Eng.	Frontend/QA
Requirements Gathering	A/R	C	C	C
System Architecture Design	A/R	C	C	C
Database Schema Design	A	R	C	I
Auth Service Development	A	R	I	I
Behavioral Data Collection	A	I	C	R
ML Model Development	A	I	R	I

Activity	PM/Lead	Backend 1	ML Eng.	Frontend/QA
ML Inference API	A	C	R	I
Admin Dashboard (React)	A	I	I	R
JavaScript Behavioral SDK	A	I	I	R
CI/CD Pipeline Setup	A	C	I	C
Unit & Integration Testing	A	C	C	R
Security / Pen. Testing	A	C	I	R
User Acceptance Testing	A/R	C	C	R
Technical Documentation	A	R	R	R
Final Presentation	A/R	C	C	C

Table 3.2 — RACI Matrix (R=Responsible, A=Accountable, C=Consulted, I=Informed)

4. Risk Assessment & Mitigation Plan

4.1 Risk Assessment Methodology

Risks are evaluated using a standard 5×5 probability-impact matrix. Each risk is assigned a Probability score (1=Very Low to 5=Very High) and an Impact score (1=Negligible to 5=Critical). The Risk Score is the product of these two values. Risks scoring 15 or above are classified as High; 8–14 as Medium; below 8 as Low.

	Impact 1 (Negligible)	Impact 2 (Minor)	Impact 3 (Moderate)	Impact 4 (Major)	Impact 5 (Critical)
Prob. 5 (Very High)	5	10	15	20	25
Prob. 4 (High)	4	8	12	16	20
Prob. 3 (Medium)	3	6	9	12	15
Prob. 2 (Low)	2	4	6	8	10
Prob. 1 (Very Low)	1	2	3	4	5

Figure 4.1 — Risk Probability-Impact Matrix (Green=Low, Amber=Medium, Red=High)

4.2 Risk Register

ID	Risk Description	Category	P	I	Score	Level	Mitigation Strategy	Owner
R-01	ML model achieves insufficient detection accuracy (TPR < 92%) due to limited or unrepresentative training data.	Technical	4	5	20	HIGH	Augment dataset with synthetic data. Use ensemble (SVM + Autoencoder). 5-fold cross-validation. Fallback to rule-based detection.	ML Engineer
R-02	High false positive rate (FPR > 5%) degrading user experience and causing abandonment.	Technical	3	4	12	MEDIUM	Tune thresholds iteratively. Implement user feedback loop. Deploy shadow mode before enforcement.	ML Engineer

ID	Risk Description	Category	P	I	Score	Level	Mitigation Strategy	Owner
R-03	Behavioral SDK causes measurable performance degradation in host app (> 5% latency increase).	Technical	3	3	9	MED	Async non-blocking event capture. Use Web Workers. Continuous performance benchmarking.	Frontend Dev
R-04	Team member unavailability due to illness or personal circumstances causing sprint delays.	Resource	3	4	12	MED	Cross-functional knowledge sharing. Thorough documentation. 2-week schedule buffer. Escalation path defined.	PM
R-05	Security vulnerability in behavioral data transmission enabling interception or replay attacks.	Security	2	5	10	MED	Enforce TLS 1.3. Short-lived JWT tokens with refresh rotation. Penetration testing in Phase 3.	Backend Dev 1
R-06	Scope creep expanding project requirements beyond agreed boundaries.	Management	3	3	9	MED	Formal change request process. Scope reviewed each sprint. Written supervisor approval for changes.	PM
R-07	Privacy compliance violation through improper storage or processing of behavioral data.	Compliance	2	5	10	MED	Store only feature vectors, never raw events. Data retention policies. Privacy impact assessment.	PM
R-08	Integration failures between microservices causing system-wide instability.	Technical	2	4	8	MED	Define API contracts (OpenAPI) before implementation. Contract testing. Circuit breaker pattern.	Backend Dev 1

Table 4.1 — Project Risk Register (P=Probability, I=Impact)

4.3 Risk Response Summary

Strategy	Applied To	Description
Mitigate	R-01, R-02, R-03, R-05, R-06, R-07, R-08	Implement preventative controls to reduce the probability or impact of the risk before it materializes.
Contingency	R-04	Pre-planned response: knowledge sharing, documentation, and schedule buffer activated upon trigger.

Table 4.2 — Risk Response Strategies

5. Key Performance Indicators (KPIs)

5.1 KPI Framework Overview

Key Performance Indicators are organized into five categories aligned with the project's strategic objectives: Security Effectiveness, System Performance, User Experience, Model Quality, and Operational Health. Each KPI includes a baseline (current state), a target value, the measurement method, and reporting frequency.

5.2 Security Effectiveness KPIs

KPI	Definition	Baseline	Target	Measurement Method	Frequency
True Positive Rate (TPR)	Percentage of genuine intrusion attempts correctly flagged as anomalous.	N/A	>= 92%	Labelled test dataset; confusion matrix analysis.	Per model release
False Positive Rate (FPR)	Percentage of legitimate user sessions incorrectly flagged as anomalous.	N/A	<= 5%	Labelled test dataset; UAT session analysis.	Monthly
Account Takeover Prevention Rate	Percentage of simulated credential-stuffing attacks detected and blocked.	N/A	>= 95%	Controlled penetration test with known-compromised credentials.	Per sec. test
Mean Time to Detect (MTTD)	Average time from anomaly onset to system detection and response trigger.	N/A	<= 60 sec	Anomaly event timestamp minus session deviation onset.	Weekly
Step-Up Auth Success Rate	Percentage of step-up challenges successfully completed by legitimate users.	N/A	>= 95%	Auth logs: step-up initiated vs. step-up passed by verified users.	Weekly

Table 5.1 — Security Effectiveness KPIs

5.3 System Performance KPIs

KPI	Definition	Baseline	Target	Measurement Method	Frequency
API Response Time (p95)	95th percentile response time for the behavioral analysis API endpoint.	N/A	<= 200ms	Prometheus metrics; Grafana dashboards; load testing with k6.	Continuous

KPI	Definition	Baseline	Target	Measurement Method	Frequency
System Uptime	Percentage of time the authentication service is available and responding.	N/A	$\geq 99.9\%$	Windows Server uptime monitor; IIS/Kestrel health checks	Continuous
Event Processing Throughput	Number of behavioral events processed per second by the Behavior Collector.	N/A	$\geq 5,000$ ev/sec	Message queue consumer metrics; throughput counters	Continuous
Concurrent Session Capacity	Max simultaneous monitored sessions without performance degradation.	N/A	$\geq 10,000$	Load testing with simulated concurrent sessions.	Per load test
DB Query Response Time	Average execution time for critical database queries (Microsoft SQL Server).	N/A	≤ 50 ms avg	SQL Server DMVs (sys.dm_exec_query_stats); Grafana dashboard	Continuous

Table 5.2 — System Performance KPIs

5.4 User Experience KPIs

KPI	Definition	Baseline	Target	Measurement Method	Frequency
False Step-Up Challenge Rate	Percentage of sessions where a legitimate user receives an unwarranted step-up challenge.	N/A	$\leq 2\%$	Session logs filtered by verified legitimate users who received challenges.	Weekly
User Adoption Rate	Percentage of application users actively protected by the behavioral monitoring system.	0%	$\geq 90\%$ in 60 days	SDK init events vs. total authenticated sessions.	Monthly
Step-Up Completion Time	Average time taken by users to complete a step-up authentication challenge.	N/A	≤ 30 sec median	Timestamps: step-up prompt issued vs. verification submitted.	Weekly
User Complaint Rate	Number of user-reported authentication issues per 1,000 active sessions.	N/A	≤ 2 per 1,000	Support ticket system; feedback form submissions.	Monthly

Table 5.3 — User Experience KPIs

5.5 ML Model Quality KPIs

KPI	Definition	Baseline	Target	Measurement Method	Frequency
AUC-ROC Score	Area under the ROC curve for the anomaly detection model.	N/A	≥ 0.95	Scikit-learn roc_auc_score on held-out validation dataset.	Per model release
Model Drift Rate	Rate at which model accuracy degrades over time as user behavior evolves.	N/A	$< 3\%$ / month	Monthly re-evaluation on live session samples vs. training baseline.	Monthly
Profile Build Time	Number of sessions required before a reliable behavioral fingerprint is established.	N/A	≤ 5 sessions	Model validation: measure FPR after N training sessions per user.	Per model release
Retraining Frequency	How often the model is retrained to incorporate new patterns and feedback.	N/A	Monthly min.	ML pipeline logs; retraining job audit records.	Monthly

Table 5.4 — ML Model Quality KPIs

5.6 Operational Health KPIs

KPI	Definition	Baseline	Target	Measurement Method	Frequency
Sprint Velocity	Average number of story points completed per 2-week sprint.	Sprint 1 baseline	$\pm 15\%$ of baseline	Jira / GitHub Projects sprint reports.	Per sprint
Defect Escape Rate	Percentage of defects discovered in production not caught during testing.	N/A	$\leq 5\%$	Bug tracker: production issues vs. total issues closed.	Monthly
Test Coverage	Percentage of codebase covered by automated unit and integration tests.	0%	$\geq 85\%$	pytest-cov / Jest coverage reports in CI pipeline.	Per CI build
Mean Time to Recovery (MTTR)	Average time to restore service following an unplanned outage or incident.	N/A	≤ 30 min	Incident logs: incident opened vs. service restored timestamps.	Per incident
Documentation Completeness	Percentage of system components with up-to-date technical documentation.	0%	100% by M10	Documentation audit against component checklist.	Monthly

Table 5.5 — Operational Health KPIs

5.7 KPI Dashboard & Reporting

All KPIs will be tracked through a combination of automated monitoring tools and manual reporting mechanisms. A consolidated KPI dashboard will be maintained in the project management tool (Jira / Confluence) and reviewed at the start of each sprint.

- Automated KPIs: Prometheus + Grafana dashboards for real-time system performance and ML metrics.
- Sprint KPIs: Reviewed in bi-weekly sprint retrospectives; trends visualized as burn-down and velocity charts.
- Monthly KPIs: Compiled into a Monthly Health Report submitted to the project supervisor.
- Milestone KPIs: Formally evaluated at each project milestone gate; results documented in milestone completion reports.
- Escalation: Any KPI breaching its target for two consecutive periods triggers a formal risk review and corrective action plan.